Circular Campus Feasibility Study

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Project

Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Executive Summary

We recommend a conditional go for the Circular Campus educational prototype. The team will assess adapting an existing asset-management system before committing to a custom application. Either route must demonstrate resource discovery, controlled transfers, asset history and every required AI function. The recommendation depends on an early technical proof, sufficient team availability and synthetic operational data. No Khalifa University (KU) staff or potential-user interviews have been conducted as of 28 September 2026, so operational adoption and local demand are not established. This recommendation does not authorize deployment with live KU records.

The original nine-week planning baseline assumes three students each contribute 8 hours per week, giving 216 person-hours against 180 hours of work and a 36-hour contingency allowance. From 28 September, eight scheduled weeks remain, equivalent to 192 hours under the same availability assumption; Section 8 distinguishes this remaining capacity from the original baseline. The proposed prototype cash ceiling is AED 300. These are planning assumptions, not approved spending or confirmed availability.

The illustrative benefit estimate is AED 12,000 in net avoided purchasing cost over 12 months, assuming 20 qualifying transfers at AED 600 each after repair and transfer costs. After the AED 9,120 first-year economic cost, the model leaves AED 2,880 and a return on investment (ROI) of 31.6%. It breaks even at 16 qualifying transfers; only 10 would produce a loss. These are calculated scenarios, not measured KU savings or a demand forecast. The base case assumes existing hosting equipment. Section 5 shows the calculation and a paid-server sensitivity that raises first-year cost to AED 11,235.36 and break-even to 19 transfers.

1 Project Scope and Objectives

Included scope and deliverables

The Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System will allow authorized departments to publish surplus assets, request suitable resources and document approvals, release and receipt. Each asset will retain its inspection, maintenance, repair and final disposition history. AI will assist users with classification, compatible-resource discovery, life-cycle recommendations, natural-language requests and reporting. People with the relevant authority will make approval and custody decisions.

The semester deliverable is an educational web prototype, a labelled test dataset, source code, a requirements specification, design documentation and recorded test results. The initial test environment will contain 500 synthetic assets distributed across 5 fictional departments and support a workload of 10 concurrent users. These values define the evaluation environment, not the size of the university's inventory or its expected demand.

Requirement group	Prototype coverage	Completion evidence
FR-01 and FR-02	Accounts, department permissions, asset registration, validation and publication	Permitted users create valid listings; prohibited changes and invalid quantities are rejected
FR-03 and FR-04	Search, requests, reservations, approvals, release and receipt	A request completes without overbooking; custody changes only after the authorized confirmations

Requirement group	Prototype coverage	Completion evidence
FR-05 and FR-06	Maintenance and life-cycle history, notifications and operational reports	Events remain traceable; notifications follow status changes; totals reconcile with stored records
AI-01 and AI-02	Classification suggestions and ranked resource matches with explanations	Users can correct suggestions; incompatible assets are excluded before ranking
AI-03, AI-04 and AI-05	Life-cycle advice, natural-language assistance and summaries of recorded activity	Advice uses available evidence; request submission requires confirmation; summaries do not invent numerical results

All five AI functions remain in scope. Their first implementation will use constrained inputs and a small evaluation dataset. Photographs and documents may accompany an asset. English text is the proposed initial input language. The feasibility study includes estimated pilot costs and benefits, but the semester deliverable remains a classroom prototype.

Excluded scope

The prototype excludes live university deployment, automated purchasing, public trading, payment processing and direct integration with KU procurement or identity systems. It will not import confidential university records or authorize real transfers, donations or disposal. Automatic interpretation of every image or document format and support for additional languages are outside the initial evaluation. AI cannot approve an exchange or make an unreviewed custody decision. Any later pilot or scope expansion requires separate authorization and a revised effort estimate.

SMART objectives and success criteria

The six criteria below retain the Phase 1 acceptance targets and identify the evidence and accountable student for each result. The internal completion target for every criterion is 22 November 2026, ahead of the final milestone week beginning 23 November. This is a team planning target, not an announced submission timestamp. Feasibility of the workload will be reviewed after the early technical proof; these targets are not results already achieved.

Objective	Pass condition and retained evidence	Task owner
Complete the exchange workflow	Pass all 10 defined workflow scenarios, including request, approval, release and receipt. Retain inputs, expected outcomes and actual results.	Saeed Alkaabi
Enforce authorization	Reject every prohibited action in the defined role and department permission matrix. Record the attempted operation and rejection.	Ali Alawadhi
Meet search response time	At least 95% of structured searches (950 of 1,000) finish within 2 seconds after warm-up, with 500 assets and 10 concurrent users. Retain timing logs.	Humaid Al Ali

Objective	Pass condition and retained evidence	Task owner
Demonstrate usability	After a 15-minute introduction, at least 4 of 5 testers independently search, submit a request and check its status. Retain task outcomes and observed errors.	Ali Alawadhi
Evaluate resource matching	At least 16 of 20 labelled feasible requests return an acceptable asset in the first 3 results. No negative eligibility case may pass the 10-case test set. Retain ranked outputs and labels.	Humaid Al Ali
Verify impact calculations	All 10 calculation cases reproduce their expected values. Retain the underlying inputs and reconciliation for each result.	Humaid Al Ali

2 Stakeholders and Operational Feasibility

Assumed current process at KU

For this feasibility study, the current-process baseline assumes that departments coordinate asset reuse through email or internal request forms, with Excel or shared spreadsheet registers for asset details and separate maintenance records. This is an assumption for validation, not a verified description of KU practice. No specific institutional software, office structure or approval chain is asserted. The following sequence identifies the records and business task owners that the team will confirm during consultation.

Process stage	Assumed tools and records	Assumed business task owner
Request a resource	Email or internal form records the specification, quantity and required date.	Department requester
Check availability	A spreadsheet register and physical check establish condition, location and available quantity.	Asset custodian
Authorize the exchange	An email decision or approval form records the request reference, decision maker and outcome.	Designated approver
Release and receive	A handover form or email confirmation records release and receipt.	Sending custodian and receiving requester
Update custody and stock	The register records the new custodian, location and remaining availability.	Asset custodian
Record upkeep and benefits	Maintenance records hold repair details; purchase and transfer records support any avoided-cost estimate.	Maintenance staff, with finance validating benefit inputs

Under this assumed arrangement, checking several records could leave availability or handover status unclear. These are possible weaknesses to investigate, not observed failure rates at KU. The proposed system would link the request, approval and custody history in one controlled record and offer a search across eligible assets. Ali will document the actual tools, forms, approval roles and a representative exchange during the Section 2 consultations, replacing any assumption that the evidence contradicts.

System functional owner

The proposed functional owner is a KU-designated Asset Management Lead responsible for the campus exchange process. This is a proposed business responsibility, not a verified KU job title; the exact office and individual require university confirmation. This owner would approve business requirements, define approval and custody rules, prioritize changes, agree acceptance criteria and sponsor any later pilot. Functional ownership does not automatically authorize every transfer; individual exchanges must still follow the designated approval roles.

The functional owner is distinct from the system administrator, who manages software access and backups, and from the student development team. Saeed, Ali and Humaid own the implementation and verification tasks assigned in Sections 1 and 8; they do not own KU assets or institutional policy. In the classroom evaluation, an instructor-approved proxy may clarify business rules, but cannot grant live data access or transfer authority.

Responsibilities and expectations

Operational feasibility depends on staff maintaining availability and confirming physical exchanges. A recommendation cannot compensate for an inaccurate condition record or an approver who has not been assigned. The proposed responsibilities below separate the administrative operation of the software from authority over university assets.

Stakeholder	Required operation and information	Proposed acceptance condition
Department requester	Describe specifications, quantity and required date; inspect matches and request status	Submit a complete request and identify its current status without assistance
Asset custodian	Maintain condition, quantity and location; publish availability and confirm release	Cannot reserve more than available stock or release a transfer without approval
Designated approver	Review request purpose and asset ownership; approve or reject with a reason	Every decision records the approver, time, affected request and outcome
Maintenance staff	Record defects, inspections, repairs, cost and resulting condition	An unserviceable asset is removed from working-resource matches
Procurement and finance staff	Review avoided-purchase assumptions and repair or transfer costs	Each financial estimate can be traced to a completed event and its inputs
Sustainability staff	Review reuse counts, asset mass and environmental assumptions	Missing mass or emissions factors produce an unavailable estimate rather than a fabricated value
System administrator	Manage accounts, roles, departments and backup access	Administrative access alone cannot approve a business transfer

The project team will implement and test these operations. The instructor and teaching assistant will assess the deliverables and clarify course expectations. Proposed technical responsibilities appear in Section 8; they do not claim that work has already been completed.

Stakeholder evidence and consultation plan

No interviews with KU staff or potential users have been conducted as of 28 September 2026. The workflow and roles above are proposed requirements derived from the project materials, not stakeholder-confirmed findings. External examples in Section 3 establish that reuse platforms exist, but cannot establish KU adoption or identify its actual approval chain.

The initial internal target of 27 September 2026 passed without any completed interviews. This is an outstanding research task, not evidence of stakeholder acceptance. Ali will coordinate the first 3 structured consultations, covering a requester, an asset custodian and an approval role, as a priority within the proposed 4 October follow-up study. If direct access is unavailable, the team will request TA approval for clearly identified proxy participants. No proxy arrangement has been confirmed. A proxy response cannot establish KU policy, authorize a transfer or be reported as a KU staff interview.

The fuller study retains the proposed sample of 10 consultations: 2 departmental requesters, 2 custodians, 2 maintenance representatives, 2 finance or procurement representatives and 2 approval or sustainability representatives. The initial 3 are included in this total where their roles match. The internal target for completing the sample and updating the requirements is 4 October 2026, ahead of the requirements milestone. These sample sizes are team planning choices, not prescribed interview counts or statistically representative samples.

Each consultation will answer six questions: Which tool or register is used now? How is a surplus asset identified and its condition checked? Who approves, releases and receives it? Which recent request illustrates a delay, duplication or unmet need? What evidence supports the replacement, repair and transfer costs? Which proposed step would the participant accept or change? The interview record will contain the date, respondent role, direct or approved-proxy status, a concise answer summary and any permitted supporting document. Unavailable records and unanswered questions will remain explicit.

Ali will map each future finding to the affected requirement ID and record whether it confirms, changes or rejects the proposed rule. Completed interviews and planned consultations will be recorded separately, including any unresolved disagreement. Humaid will check the benefit inputs against any cost evidence obtained. This submission contains no interview-derived findings; the operational conclusion remains conditional, and documented follow-up findings will inform the requirements revision.

The proposed local-demand threshold requires at least 6 of 10 distinct direct KU respondents to report a relevant recurring need, together with confirmation from authorized representatives of at least 3 departments that each can identify 10 potentially reusable assets. The resulting 30 candidate assets would support a small pilot discussion, not a campus-wide demand estimate. Direct responses and approved-proxy responses must be counted separately. If fewer than 10 direct respondents participate, report the actual denominator and leave this threshold unassessed. Proxy feedback can test the proposed workflow but cannot establish KU demand or departmental commitments.

Before a pilot, 5 testers will receive a 15-minute introduction and then independently search for an asset, submit a request and check its status. At least 4 must complete all three tasks without help. Any failure will identify the screen, misunderstood field or missing instruction requiring correction. One custodian and one approver must be assigned for each participating department, and at least 1 test exchange must complete approval, release and receipt before real exchanges are considered.

Operational feasibility is not yet demonstrated. The defined workflow provides a basis for interviews and prototype tests, but a consultation plan alone does not establish staff participation, approval ownership or user

acceptance. The conditional prototype recommendation can be reassessed once documented findings and observed task results are available.

3 Market Analysis

Demand and existing alternatives

The relevant market is internal university resource reuse, rather than public sales of second-hand goods. The immediate users are departments holding surplus assets and departments with compatible needs. Local demand requires validation through the consultations in Section 2; the existence of unused equipment or avoidable expenditure at this university has not been quantified.

There is evidence that institutional reuse systems serve a real need. UCL's October 2020 account of its Warp-it platform reports total savings of GBP 0.5 million during its first six years of use [1]. This historical example supports the practicality of departmental reuse, but its scale and savings cannot be transferred to this project as a forecast.

Alternative	Established capability or proposed approach	Implication for this project
Spreadsheet and email workflow	A manually maintained register and messages can record assets and requests	Low setup effort, but reservations, role checks and linked approval history would require manual controls
Snipe-IT	Its documented functions include requestable assets, check-in and checkout, permissions, asset history and maintenance records [2]	A credible asset-management alternative; a configured trial would be needed to assess the complete course workflow and AI evaluation requirements
Warp-it	UCL describes staff using Warp-it to donate and claim surplus reusable items within the university [1]	A direct reuse alternative; procurement terms and fit with the university's approval model require assessment
Proposed Circular Campus prototype	An exchange prototype, implemented through adaptation or custom development, with measured comparisons between conventional search and AI assistance	Supports documented requirements, design decisions and testing; the selected route must identify the team's own changes and development effort

The comparison does not claim that competitors lack a feature simply because it is absent from a public description. The market evidence establishes that asset management and institutional reuse systems already exist; it does not measure KU demand or prove that a new system is preferable. Platform selection will consider approval and custody fit, permission control, AI evaluation, licensing, integration effort and ongoing maintenance. A configured trial must resolve those questions before the team commits to an implementation route.

Adapting an existing system

Snipe-IT is the named candidate for an adaptation trial. Alongside the asset requests and history described above, its product documentation lists custom fields and a JSON REST API [2]. These documented capabilities provide a possible starting point for extension; they do not prove that the complete course workflow or AI requirements are already satisfied.

The proposed improvement would retain suitable asset records and existing controls, then configure or extend the system where the trial identifies a gap in departmental approval, custody confirmation, eligibility-aware matching or impact reporting. All five AI functions and the comparison with conventional functionality remain required. A requirements map will classify each FR-01 to FR-06 and AI-01 to AI-05 item as reusable, configurable, requiring extension or unmet. Existing features will not be presented as work developed by the team.

By the internal planning target of 4 October 2026, Saeed will coordinate the platform decision with Ali assessing workflow and permissions and Humaid assessing AI access and reporting. The decision record must include the requirements map, licence obligations, setup and integration constraints, and estimated remaining development and maintenance effort. Adaptation will be selected only if the required controls can be demonstrated, all required functions have a feasible implementation path and its estimated total effort is lower than the custom alternative. Otherwise the team will retain the custom reference design or revise the plan with the instructor. This comparison and the 12-hour technical proof are included in the first 40-hour work period, not added to the 216-hour capacity. No adaptation saving is assumed before that assessment.

Market feasibility conclusion

The category is established, but local adoption and purchasing intent are unproven. The recommended next step is a bounded educational prototype and demand validation, not a claim that a commercial launch or university purchase is justified. The prototype should demonstrate permission-aware matching and traceable decisions rather than rely on an unsupported claim of uniqueness.

4 Technical and Data Feasibility

Proposed implementation approach

For the custom-development alternative, the reference stack is Python with Django, server-rendered HTML and CSS, PostgreSQL, and a locally executed sentence-embedding model. The team would implement accounts, forms and administration in Django and test department-level permissions and business approvals. PostgreSQL would store assets, requests, reservations, decisions, transfers, maintenance events and audit records. If adaptation is selected, the team will use the existing platform's supported configuration and interfaces; the Django and PostgreSQL design is not assumed to describe that platform. The technical proof will confirm the selected route, skills and operating environment.

The custom alternative uses a single web application rather than separately deployed front-end and back-end services to limit interfaces and deployment components. Either implementation route must protect a transfer so custody, location and history change consistently. Reservation checks must also prevent two users from allocating the same remaining quantity. An adapted implementation must demonstrate those controls through its supported interfaces and tests rather than bypass them.

Component	Proposed resource or method	Feasibility check
Development and demonstration	Existing computers; target 4 CPU cores, 8 GB RAM and 10 GB free storage; one machine hosts the local demonstration	Record actual specifications and run the representative workload; these values are planning targets, not measured minimum requirements
Application and database	Django and PostgreSQL for the custom route, or the selected platform configuration; local test accounts and department-scoped records	Two users cannot reserve the same final item; interrupted receipt leaves a consistent database
Semantic matching	Sentence Transformers all-MiniLM-L6-v2 with precomputed asset embeddings [3]	Filter mandatory specifications first; evaluate ranking on a fixed labelled request set
Classification and metadata	Compare asset text with defined category descriptions; present suggested labels for review	Custodian can accept, correct or reject each suggestion; required fields are never silently invented
Assistant and generated reports	One text-generation service behind a replaceable interface, using approved synthetic or non-sensitive inputs	Verify service terms, response validity and the AED 150 experimental allowance before selecting a provider
Operational reporting	Database queries calculate counts and totals; generated text explains those supplied results	Reconcile every reported number with a query result or recorded calculation
Files and backup	File-type and size checks, restricted storage, daily backup during active work and one restore test	Reject unsupported files and restore both records and their linked files

The proposed matching method will compare request embeddings with precomputed asset-description embeddings using cosine similarity. Its suitability for campus assets must be evaluated against the same labelled cases as conventional search. The selected model and eligibility rules remain subject to this evaluation.

No dedicated GPU purchase is budgeted. If local inference cannot meet the required workload, the team will benchmark a smaller or optimized model before considering paid compute. A text-generation service remains a dependency: a canned report is a fallback for continuity, not evidence that the generative-AI requirement has been implemented.

Capacity and skills

The proposed model produces 384 values per asset, and its documentation states that inputs longer than 256 word pieces are truncated by default [3]. Word pieces are model tokens, not necessarily complete words. Storing 500 embeddings as 32-bit values requires 500 × 384 × 4 = 768,000 bytes, approximately 0.73 MiB, before indexing and application overhead. This is the team's storage calculation, not a measured total memory requirement. Essential specifications will be stored in structured fields and checked separately so they cannot be lost through text truncation.

For planning, one photograph capped at 1 MB and one document capped at 2 MB per asset could consume up to 1.5 GB for 500 assets, before backups, history and software. The 10 GB storage allowance must be checked

against the installed environment and retained backup copies. Larger uploads require a revised storage estimate.

Required skills for the custom route are Python web development, HTML forms, relational data modelling, Git collaboration, authorization testing and embedding-based retrieval. Adaptation additionally requires knowledge of the candidate platform's configuration, interfaces and extension mechanisms. Current proficiency is not assumed. The technical proof has a 12-team-hour limit within the first 40-hour work period and an internal target of 4 October 2026. On the candidate route it must demonstrate login, a valid listing, a departmental request, protected approval, release and receipt, plus ranked search over 20 sample assets. Failure to complete the proof within 12 hours will trigger a revised estimate before committing to the platform.

The team may use Codex for coding assistance, as permitted by the TA. Each member must be able to explain the purpose, inputs, outputs, permission checks and error handling of code they contribute, and demonstrate its tests. Generated code will be reviewed in small changes before it is committed. AI assistance does not establish technical proficiency or replace the early proof and recorded test results.

Data quality and AI evaluation

The 500 sample assets will include five categories: furniture, computers and monitors, laboratory equipment, books, and general supplies. Test records will use fictional people, departments and locations. All 500 must pass checks for unique identifiers, valid status, positive quantities and required publication fields. Units and categorical values will be normalized so, for example, an HDMI requirement cannot be satisfied solely by similar wording.

The ranking benchmark will contain 20 feasible requests with acceptable assets labelled before evaluation. These requests will not be used to tune ranking weights. Conventional keyword or filter search and semantic ranking will use the same records and labels. The target is at least 16 requests with an acceptable item among the first 3 results. A score of 16/20 is an 80% success rate on that test set, not evidence of 80% general AI accuracy.

An additional 10 negative cases will include unavailable assets, insufficient quantity, incompatible specifications and unacceptable condition. None may pass the eligibility filter. Asset descriptions and uploaded text will be treated as data, not instructions capable of changing permissions or issuing approvals. The assistant will display a proposed request for user confirmation and will not authorize transfers or confirm physical receipt.

OWASP identifies prompt injection as a risk when applications mix untrusted text with model instructions, and recommends separation, least privilege and human oversight [4]. For this prototype, model output will not directly execute database updates. The application will independently validate the requested operation and the user's permission before any confirmed change. Five additional security tests will place instruction-like text in asset descriptions or assistant inputs and check that it cannot bypass approval, expose another department's protected records or alter custody. These tests supplement, rather than prove the completeness of, the security controls.

Structured-search performance will be measured over 1,000 requests at 10 concurrent users after a documented warm-up, with at least 950 completing within 2 seconds. Usability, workflow and impact-calculation tests will retain the Phase 1 sample sizes. Numerical summaries will come from calculations outside the language model. Technical feasibility remains conditional until these tests and the early technical exercise produce evidence.

5 Financial Analysis

Prototype cash and effort budget

The cost estimate states scope and assumptions, separates cash expenditure from imputed labour, and tests sensitivity to transfer volume, unit benefit and additional hosting. The 180-hour estimate remains the provisional custom-development baseline. An adaptation decision requires a revised estimate covering configuration, extension, integration and maintenance; no saving from reused software is credited yet. Team hours, the labour valuation, transaction benefits and service allowances remain project assumptions until validated.

The following amounts are proposed allowances in UAE dirhams, not supplier quotations, actual expenditure or an approved budget. Existing computers and internet access are assumed available. No new hardware or institutional integration is included. The team's time is counted separately so a low cash requirement is not mistaken for a cost-free project.

Prototype item	Proposed cash allowance in AED	Basis
Existing computers and local open-source application tools	0 incremental purchase	Reuse existing equipment; no dedicated GPU or paid database licence
Text-generation experiments	150	Spending ceiling across the development period; provider selection depends on fitting this limit
Optional short-term deployment or service testing	100	Allowance if a local demonstration is insufficient; not a quoted hosting plan
Backup media or incidental technical costs	50	Contingency allowance
Total prototype cash ceiling	300	Expenditure requires team agreement before purchase

Baseline development effort is 180 person-hours. At an illustrative valuation of AED 40 per hour, its economic value is AED 7,200. The hourly value is a sensitivity assumption, not a student wage or market salary. The 36-hour schedule reserve is outside the baseline; using all of it adds AED 1,440 to the imputed development cost. Actual cash payments for student time are not assumed.

Illustrative first year pilot economics

This scenario asks whether reuse benefits in the first 12 months of pilot operation could cover upfront development and that year's running costs. It is separate from the semester cash budget and does not establish production funding. The base case assumes local hosting on existing equipment at no additional server charge; it excludes a new production server, institutional integration and extensive support. Such additions would require a new business case.

Cost component	Assumption	First year economic cost in AED
Development time	180 hours at AED 40 per hour	7,200
Initial prototype cash allowance	Full ceiling treated as spent	300
Pilot AI and ancillary services	AED 50 per month for 12 months; existing host	600
Pilot backup allowance	AED 5 per month for 12 months	60
Ongoing maintenance time	2 hours per month for 12 months at AED 40 per hour	960
Total	Development, initial cash and one year of operation	9,120

The annual operating allowance is AED 1,620: AED 660 in services and backup plus AED 960 in maintenance effort. Including the initial AED 300, the first-year cash allowance is AED 960; the remaining AED 8,160 is imputed labour. The 24 pilot maintenance hours are additional to the semester development capacity. Two maintenance hours per month are a limited pilot assumption covering dependency checks, backup review and minor support. Actual support demand may be higher. Monetary benefits will be counted only where a completed reuse event avoids a purchase that would otherwise occur.

For the illustrative unit benefit, the avoided replacement cost is AED 800, repair costs are AED 150 and transfer costs are AED 50. The net avoided cost is therefore AED 600 per qualifying event. Repair and transfer costs are deducted here and are not deducted again in the operating budget. A transfer without an avoided purchase may have reuse value, but it does not receive this financial benefit automatically.

Estimated benefits ROI and sensitivity

The illustrative benefit estimate is AED 12,000 over 12 months: 20 qualifying transfers multiplied by (AED 800 avoided replacement cost minus AED 150 repair cost minus AED 50 transfer cost). This is net of those event costs but before the AED 9,120 development and operating cost. The resulting first-year surplus is AED 2,880. The inputs and transfer count are assumptions, not KU observations or supplier quotations. Better custody traceability and easier discovery are additional intended benefits; no percentage time saving or environmental reduction is claimed without a measured baseline.

First-year ROI is calculated as the annual net avoided cost minus AED 9,120, divided by AED 9,120 and expressed as a percentage. The denominator includes imputed development and maintenance time. Environmental benefits and staff time savings are not monetized in this model.

Annual qualifying transfers	Net avoided cost in AED	First year surplus or loss in AED	First year ROI
10	6,000	-3,120	-34.2%
20	12,000	2,880	31.6%
30	18,000	8,880	97.4%

At AED 600 per event, 15 transfers provide AED 9,000 and fall short of the AED 9,120 cost. A minimum of 16 qualifying transfers is therefore required for first-year break-even. If the net benefit is only AED 300 per event, the threshold rises to 31 transfers. If all 36 reserve hours are used, the first-year cost rises to AED 10,560, the 20-transfer ROI falls to 13.6%, and break-even requires 18 transfers.

Published hosting benchmark

As checked on 28 September 2026, DigitalOcean lists a Basic Regular shared-CPU server with 8 GiB memory, 4 vCPUs and 160 GiB SSD at USD 48 per month [5]. This is a price benchmark, not a selected supplier or proof of equivalent performance to the proposed local computer. Using AED 3.6725 per USD from the Central Bank of the UAE's published exchange-rate table [6], the benchmark is AED 176.28 per month or AED 2,115.36 for 12 months. The conversion excludes payment fees, taxes, separately charged services and any additional support.

If that server is added while retaining the base case's AED 600 annual AI and ancillary-service allowance and other costs, the first-year economic cost rises from AED 9,120 to AED 11,235.36. At 20 qualifying transfers, the surplus falls to AED 764.64 and ROI to 6.8%; break-even requires 19 transfers at AED 600 each. These are calculated sensitivities, not a complete production quotation. The paid-server cost alone exceeds the AED 50 monthly service allowance. A live pilot requires a fresh hosting, backup and support budget.

The financial conclusion is conditional: a small prototype can be attempted within a limited cash ceiling, but a pilot's economic case depends on validated avoided purchases, unit costs and support requirements. A positive scenario is not a forecast, and no campus savings have been achieved by this project yet.

6 Legal Regulatory and Ethical Analysis

Privacy and data authorization

Names, account identifiers, staff responsibilities and request histories can be personal or confidential information. The prototype will use fictional operational data and will not import university accounts, purchase records or asset files without authorization. Real personal records, credentials and service keys must not enter the shared source repository or an external AI prompt.

The UAE Personal Data Protection Law, Federal Decree-Law No. 45 of 2021, is relevant to a privacy review, but its scope contains exclusions, including government data and processing by governmental entities under Article 2 [7]. The applicable regime cannot be determined solely from the project's university setting. Before a live pilot, the university's authorized legal and data-governance staff must confirm applicable rules, permitted purposes, retention periods, access rights and any cross-border processing. This study is a feasibility screening, not a finding of legal compliance.

Proposed controls are minimum-necessary fields, role and department restrictions, separate credentials, encrypted network connections for remote access and a documented process for deleting or correcting

authorized personal records. Any external AI service requires review of its data retention, training use and processing location before real data are transmitted. Articles 5, 15 and 22-23 address processing controls, correction or erasure and cross-border transfers respectively [7]; their applicability must be confirmed before deployment. Until that review, AI experiments will use synthetic inputs only.

Intellectual property ownership and safety

The IP review will cover source code, third-party libraries, model licences, photographs and uploaded documents. The UAE copyright framework includes Federal Decree-Law No. 38 of 2021 [8]. The team will retain applicable licence notices and use only images or documents it owns or has permission to use. An open-source dependency does not make every accompanying dataset or uploaded image unrestricted.

Constraint	Project control	Condition before a live pilot
Asset ownership and procurement authority	Separate software access from permission to transfer, donate or dispose	Written approval rules and named institutional decision makers
Software and model licensing	Maintain a dependency and licence record; check the selected model and service terms	All distributed components and inputs have compatible permissions
Laboratory and electrical safety	Block unserviceable assets from working-resource matches; require qualified inspection where applicable	Safety review before an asset is released for physical use
AI accountability	Show supporting factors and uncertainty; require human decisions	No automatic disposal, approval or unreviewed change of custody
Security and confidentiality	Test permission boundaries and review repository contents	No unresolved critical access defect or exposed credential

Legal and ethical feasibility is acceptable for a synthetic-data classroom demonstration under these controls. Institutional deployment remains conditional on university authorization and specialist review.

7 Risk Assessment and Contingencies

The team will assign an owner to each risk and define an observable trigger and contingency. Matching and access tests will check unsafe recommendations and disclosure scenarios. Failed functions will be corrected or disabled before release, and owners will review unresolved risks against the next milestone. The numerical ratings below are the team's planning method.

The following scores are planning judgements, not statistical probabilities. Likelihood is rated 1 for unlikely, 2 for plausible and 3 for likely. Impact is rated 1 for a small rework task, 2 for a threatened milestone and 3 for loss of required scope or unsafe data handling. The priority score is likelihood multiplied by impact. A score of 6 or 9 requires action before the affected milestone.

Risk and initial score	Observable trigger	Proposed owner	Contingency
Learning and integration effort exceeds plan 3 × 3 = 9	The early end-to-end exercise exceeds its allocated effort or cannot run on the chosen computer	Saeed	Pair on the failing component, use reserve hours and revise the estimate before fixing the stack
Insufficient team availability 2 × 3 = 6	Remaining capacity falls below remaining work plus a realistic defect allowance	All members	Reallocate tasks and simplify presentation or infrastructure work; seek approval for any change to required feature depth
Stakeholder research remains incomplete 3 × 2 = 6	The 27 September initial target passed; no interviews were completed as of 28 September	Ali	Arrange direct consultations or seek TA approval for proxies; retain synthetic records and the conditional conclusion until evidence is obtained
AI recommendations are unsuitable 2 × 3 = 6	Fewer than 16 of 20 requests meet the top-three target, or any negative eligibility case passes	Humaid	Fix mandatory filters, examine labels and compare alternatives without removing the baseline search
API cost or service failure 2 × 2 = 4	Experimental spending reaches AED 120 of the AED 150 allowance, or the service fails	Humaid	Review usage before the ceiling; retain conventional functions and assess a compliant replacement
Unauthorized disclosure or access 2 × 3 = 6	Any protected operation succeeds for an unauthorized role, or real confidential data appear in a test input	Ali	Stop the affected operation, remove the exposure and retest before release
Data inconsistency or overbooking 2 × 3 = 6	Two concurrent requests reserve the same final item, or an interrupted transfer changes only part of the record	Saeed	Correct transaction handling and rerun concurrent and failure tests
Benefits are overstated 2 × 2 = 4	A claimed saving lacks an avoided-purchase basis or a cost input	Humaid	Exclude the unsupported estimate and recalculate ROI from validated events

Owners and ratings are proposed. The stakeholder-research gap is already present; the other triggers describe risks to monitor. Risks will be reviewed weekly against actual hours, defects and the next deliverable. A failed privacy or authorization check prevents a live pilot regardless of the total risk score.

8 Project Schedule and Resources

Capacity and milestones

The milestone baseline follows slide 2 of the supplied Phase 2 feasibility-study presentation, as directed by the TA. The later one-week Phase 2 extension is applied separately. The older tentative timetable in the full project brief is not the scheduling baseline. The slide milestones specify lab weeks; the confirmed online Phase 2 deadline is stated below. Capacity and hour allocations remain team assumptions.

The slide schedule places Phase 2 in the week of 21 September 2026, Phase 3 requirements in the week of 5 October, Phases 4 and 5 design in the week of 26 October, Phase 6 draft implementation in the week of 16 November, and Phases 7 and 8 testing, final submission and demonstration in the week of 23 November. Following the separately announced one-week extension, Phase 2 moves to the lab week of 28 September. The confirmed Blackboard submission deadline is Monday, 28 September 2026 at 6:00 p.m. UAE time. The extension does not by itself shift later milestones or add development capacity.

The original capacity baseline covers 21 September through 22 November: 3 students × 8 hours × 9 weeks = 216 person-hours. Baseline work totals 180 hours, with a planned 36-hour reserve equal to 20% of baseline effort and 16.7% of total capacity. The estimate excludes Phase 1 work already completed. The table below preserves these original allocations; it is not a record of hours worked.

As of 28 September, eight scheduled weeks remain through 22 November, giving 192 person-hours at 24 team-hours per week. Actual effort and reserve use during 21-27 September have not been reconciled, so neither the remaining workload nor the unused reserve can be stated as a verified figure. The team must compare completed work with the original allocations and confirm availability before treating the plan as achievable.

Work period	Baseline team hours	Deliverable and dependency
21 September to 4 October	40	Revise Phase 2, validate assumptions, compare adaptation with custom development, prepare sample data and complete the technical proof and requirements specification
5 to 25 October	60	Complete architecture, data model, permission rules and detailed design; develop component proofs before full implementation
26 October to 15 November	60	Integrate the exchange workflow, maintenance, notifications, reports and all AI functions; execute tests during development
16 to 22 November	20	Run final regression and evaluation cases, correct defects, finish documentation and prepare the demonstration
Total baseline	180	Final milestone in the week of 23 November

The original baseline averages 20 team-hours per week within an assumed 24-hour weekly capacity, providing a planned margin of 4 hours weekly across nine weeks. This describes the original 36-hour allowance, not a confirmed remaining reserve. Testing occurs throughout development; the final week is reserved for regression and completion, not the first discovery of integration failures.

Development task owners

Member	Primary work	Planned baseline contribution
SAEED ALKAABI 100063100	Platform decision and application integration; asset and transfer workflow; repository coordination; ownership of the 10 workflow test cases	60 person-hours
ALI ALAWADHI 100063186	KU process validation and stakeholder requirements; data model, permissions and forms; ownership of permission and usability tests	60 person-hours
HUMAID AL ALI 100062914	Matching and AI interface; benefit and reporting calculations; ownership of search performance, matching and calculation tests	60 person-hours

Each named student is accountable for completing the assigned task and retaining its evidence, with another member reviewing major changes. The original allocation is 60 baseline hours and 12 reserve hours per member; these are not remaining-hour balances. Responsibilities may be redistributed after the skills exercise, but every task must retain a named owner. All three members must understand the complete workflow. These allocations are proposed responsibilities, not claims of completed contributions or confirmed availability.

The critical dependency sequence is validated scope, then data and permission definitions, then a working request-to-receipt exchange, followed by reliable AI integration and final evaluation. The system cannot be declared complete while a required function is replaced by a screenshot or an unimplemented placeholder. If effort exceeds capacity, the team must revise the delivery plan with the instructor rather than silently remove required functions.

9 Recommendation and Implementation Action Plan

Feasibility decision

The recommendation is a conditional go for an educational prototype, with an adaptation assessment before committing to custom development. The preferred route will be the one that can satisfy the full requirements and acceptance tests within validated effort and licence constraints. The 500-asset dataset permits a bounded AI evaluation, but technical suitability, the 180-hour baseline and local demand remain unverified. Neither route authorizes institutional deployment.

Financially, the AED 300 cash ceiling makes the classroom implementation a limited expenditure proposal. The hypothetical pilot breaks even at 16 qualifying transfers under the stated cost assumptions, but loses money in the 10-transfer scenario. Adding the published paid-server benchmark raises the break-even point to 19 transfers and reduces the 20-transfer ROI to 6.8%. This sensitivity prevents the study from treating reuse benefits as guaranteed income or assuming that free software eliminates maintenance cost.

Decision point	Required evidence	Action if the condition is not met
Phase 2 evidence status on 28 September	No KU staff or potential-user interviews completed. The current workflow, institutional roles and local demand remain assumptions.	Retain the conditional operational conclusion and complete follow-up research; request TA guidance if an approved alternative is needed
Before selecting the implementation route	Adaptation versus custom requirements map and effort comparison; login-to-transfer proof; ranked search over 20 assets; recorded hardware and team availability	Revise the platform choice, effort estimate or assignment before further implementation
Before the design milestone	Defined approval and reservation rules, complete data fields, test cases and reviewed service terms	Resolve the specific dependency or keep the affected work on synthetic inputs
Before final submission	All required capabilities demonstrated; 10 workflow and 10 calculation cases pass; permission, search, usability and matching targets evaluated	Correct defects and report any unmet target honestly
Before any live university pilot	Data and safety authorization, named operational owners, validated demand, cost inputs and hosting arrangements	Continue only the educational demonstration; do not import live records or release assets

The immediate priority is to address the missed initial consultation target and record findings against the affected requirements. The proposed follow-up target remains 4 October; no completed interviews are claimed in this submission. The team will reconcile actual effort and availability, compare adaptation with custom development, and complete the candidate route's technical proof before finalizing requirements and permission rules. It will then establish a complete request-to-receipt exchange and integrate the remaining AI and reporting functions. Named task owners will retain evidence against the success criteria. A live university pilot remains a separate decision requiring a confirmed functional owner, authorization and validated economics.

AI assistance in Phase 2

We used OpenAI Codex to assist with drafting substantial portions of this feasibility report, organizing the analysis, identifying and checking candidate external sources, checking calculations, formatting IEEE citations and preparing presentation material. AI also supported revisions following the lab feedback. The cited publications, rather than AI-generated text, support the external factual claims. AI assistance does not establish completed stakeholder consultation, measured KU benefits, tested system performance or university approval. The three authors remain responsible for reviewing the sources and assumptions, understanding the analysis and approving the final submission.

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